ModelIt!

Teacher's Guide

The Ultra-Processed Food Problem

Modeling Food Chemistry, Metabolic Pathways, and Nutritional Systems

Quick Reference

Grade Level	Grade 12L1
NGSS Standard	HS-LS1-7
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-7

Use a model to illustrate that cellular respiration is a chemical process whereby the bonds of food molecules and oxygen molecules are broken and the bonds in new compounds are formed, resulting in a net transfer of energy.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop computational models to predict how food processing level affects metabolic responses, satiety signaling, and long-term health outcomes.
Disciplinary Core Idea	**LS1.C Organization for Matter and Energy Flow in Organisms** — Cellular respiration converts food molecules into energy, but the body's metabolic response differs dramatically based on food processing level, affecting how efficiently energy is regulated.
Crosscutting Concept	**Cause and Effect** — Students identify causal mechanisms linking ultra-processing to disrupted satiety, metabolic dysfunction, and caloric overconsumption through specific biochemical pathways.

CAST Testing Alignment

MULTIPLE CHOICE:

Two meals contain identical calories, protein, fat, carbohydrates, sugar, and fiber. One is made from whole foods, the other from ultra-processed ingredients. The model predicts that the ultra-processed meal will result in greater caloric overconsumption. What mechanism best explains this?

CONSTRUCTED RESPONSE:

Using your model, explain why replacing a school's ultra-processed lunch menu with whole-food alternatives could improve student academic performance. Connect food processing to at least three components in your model.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

In the last 50 years, ultra-processed foods have gone from a small fraction of the American diet to over 60% of total calories consumed. This industrial transformation of food has coincided with unprecedented rates of obesity, type 2 diabetes, and metabolic disease. The science is now clear: it's not just what nutrients are in our food that matters, but how that food is processed.

The NOVA Classification System

Developed by researchers at the University of Sao Paulo, the NOVA system classifies foods into four groups: unprocessed or minimally processed (fruits, vegetables, meat, eggs), processed culinary ingredients (oil, butter, sugar, salt), processed foods (canned vegetables, cheese, fresh bread), and ultra-processed foods (soft drinks, packaged snacks, instant noodles, chicken nuggets). Ultra-processed foods are defined by the presence of industrial ingredients and additives not used in home cooking.

The NIH Ultra-Processing Study

In 2019, a landmark randomized controlled trial at the National Institutes of Health provided the first direct evidence that ultra-processing itself — not just nutrient content — drives overconsumption. Twenty adults lived in a research facility for four weeks, eating either ultra-processed or unprocessed meals for two weeks each. Diets were matched for calories, sugar, fat, fiber, and macronutrients. On the ultra-processed diet, participants ate 500 more calories per day and gained an average of 2 pounds in two weeks. On unprocessed food, they lost 2 pounds.

How Ultra-Processing Disrupts Satiety

Ultra-processed foods undermine satiety through multiple mechanisms: they are softer and require less chewing (increasing eating speed 20-30%), they have higher caloric density per bite, their refined ingredients are absorbed faster (causing rapid blood sugar spikes and crashes), and their additives may directly alter gut hormone signaling. The brain's appetite regulation system evolved for whole foods that required significant chewing and digestion time — ultra-processed foods bypass these ancient controls.

The Economics of Ultra-Processing

Ultra-processed foods are cheap because they use low-cost industrial ingredients (corn syrup, hydrogenated oils, starches) that have long shelf lives and require minimal preparation. Whole foods are more expensive, perishable, and require cooking skills and time. This creates a nutrition inequality where the cheapest available calories are the most metabolically harmful. In the US, households spending less than $4 per person per day on food get over 80% of their calories from ultra-processed sources.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify processing level, sugar/additive load, satiety response, metabolic health, and caloric overconsumption as the key components of the food-health system.

E **Establish Relationships**
Students determine that processing level drives additive load, additives weaken satiety, weak satiety causes overconsumption, and overconsumption degrades metabolic health in a reinforcing cycle.

V **Visualize & Model**
Students build a computational model showing how food processing creates a cascade from engineered palatability to metabolic dysfunction.

E **Evaluate Outcomes**
Students run whole food, ultra-processed, and transition scenarios to quantify how processing level affects health outcomes independent of nutrient content.

R **Revise & Extend**
Students add gut microbiome, food marketing, or socioeconomic access to explore how systemic factors amplify or mitigate the effects of ultra-processing.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Today we're investigating something that affects every single one of you. Modeling Food Chemistry and Metabolic Health."*

Watch For

Show cover image. Ask: What do you already know about this topic? Quick share-out.

Slide 2: Learning Objectives

3 min

Say: *"Here's what you'll be able to do by the end of today's lesson."*

Watch For

Have students read objectives aloud. Pre-teach vocabulary with quick visual aids.

Slide 3: Big Question

3 min

Say: *"This is our driving question. By the end of class, you'll be able to answer it with evidence from your model."*

Watch For

Quick-write: Students write their initial hypothesis. Save to compare later.

Slide 4: LEVER Framework

2 min

Say: *"We're going to build a computational model to investigate this system. Here are our five steps."*

Watch For

Preview each LEVER step. Emphasize that modeling reveals hidden patterns.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key parts of this system? Which ones come from outside, and which respond inside?"*

Watch For

Guide sorting of external vs. internal components. Use think-pair-share for reasoning.

Slide 6: Activity 2: Connections

8 min

Say: *"NIH study: same calories, same nutrients, but 500 MORE calories consumed from ultra-processed meals. Why?"*

Watch For

Students draw arrows showing +/- relationships. Discuss feedback loops if present.

Slide 7: Activity 3: Simulation

10 min

Say: *"Now let's test our model. Run each scenario and record what happens."*

Watch For

Students run all scenarios, record predictions vs. observations. Discuss surprises.

Slide 8: Discoveries

5 min

Say: *"So what did our model reveal? Ultra-processed foods bypass your body's natural fullness systems...."*

Watch For

Class discussion of discoveries. Compare initial hypotheses to model evidence.

Slide 9: STEM Challenge

12 min

Say: *"Now apply what you've learned. Design meals as satisfying as processed ones using whole foods within budget...."*

Watch For

Groups design solutions using model data. Present and evaluate designs.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Processing Level
- **INTERNAL:** Sugar and Additive Load, Satiety Response, Metabolic Health, Caloric Overconsumption

Relationships

- **Processing Level to Sugar and Additive Load:** POSITIVE (+) — Higher processing adds more industrial ingredients — sugars, emulsifiers, flavors, and preservatives that don't exist in whole foods.
- **Sugar and Additive Load to Satiety Response:** NEGATIVE (-) — Higher additive loads disrupt gut hormone signaling, accelerate eating speed, and reduce the body's ability to detect fullness.
- **Satiety Response to Caloric Overconsumption:** NEGATIVE (-) — Weaker satiety signals mean the brain doesn't receive the 'stop eating' message, leading to consumption of 20-30% more calories.
- **Caloric Overconsumption to Metabolic Health:** NEGATIVE (-) — Chronic excess calorie intake, especially from refined sources, drives insulin resistance, inflammation, and fat accumulation — the hallmarks of metabolic syndrome.

Key Discoveries

- Ultra-processed foods override normal satiety signaling, causing people to eat 500+ extra calories per day without feeling more full — it's not about willpower, it's about food engineering
- The speed of eating matters: ultra-processed foods are consumed 20-30% faster, which doesn't allow time for gut hormones to signal fullness to the brain
- Metabolic damage from ultra-processed diets is partially reversible — shifting to whole foods shows measurable improvements in blood sugar, inflammation, and gut microbiome diversity within 2-4 weeks
- Food processing level predicts health outcomes better than individual nutrients — it's not just about sugar or fat content but about the entire matrix of processing that disrupts metabolic regulation

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Processed food is fine as long as you watch calories"	The NIH study demonstrated that processing level affects health independently of calorie content. Participants given unlimited access to either diet consumed 500 more calories daily from ultra-processed food — the processing disrupted satiety signals that normally regulate intake. Calorie labels don't capture how processing changes the body's metabolic response to food.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"If the FDA allows it, it must be safe"	FDA approval means a food additive is Generally Recognized as Safe (GRAS) in isolation, not that a diet dominated by ultra-processed foods is safe. Many additives are tested individually but consumed in combinations never studied. The health effects of ultra-processing emerge from the total dietary pattern, not from any single ingredient.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Organic ultra-processed food is healthy"	Organic ultra-processed food (organic chips, organic soda, organic candy) is still ultra-processed. The NOVA classification is about the degree of industrial processing, not about whether ingredients are organic. An organic cookie can have the same metabolic impact as a conventional one — the processing is what matters, not the farming method.